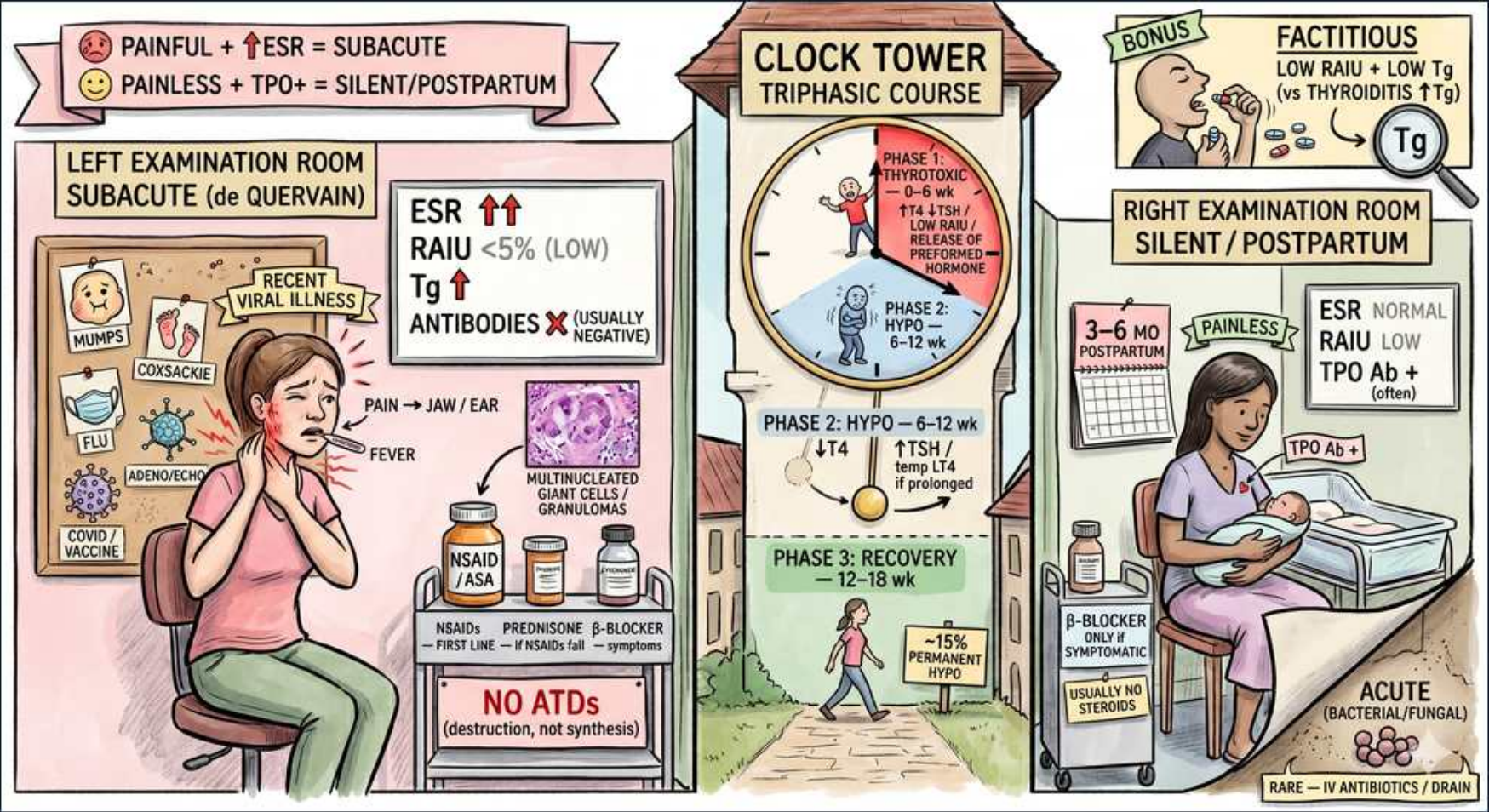


Thyroid — Painful & Silent Thyroiditis



1. Painful + ↑ESR = subacute

WHAT YOU SEE

'PAINFUL + ↑ESR = SUBACUTE'

WHAT IT ENCODES

A tender, painful thyroid with a markedly elevated ESR (and CRP) defines subacute granulomatous thyroiditis (de Quervain's), the classic post-viral destructive thyroiditis. It is the prototype of PAINFUL thyroiditis — pain + ESR is the board pairing that separates it from the painless (silent/postpartum) forms.

2. Painless + TPO+ = silent

WHAT YOU SEE

'PAINLESS + TPO+ = SILENT/POSTPARTUM'

WHAT IT ENCODES

A painless (non-tender) destructive thyroiditis with positive anti-TPO antibodies is silent (sporadic) or postpartum thyroiditis — autoimmune lymphocytic thyroiditis on the Hashimoto's spectrum. Same low-uptake destructive picture as de Quervain but no pain and antibody-positive. Mnemonic: pain → de Quervain (ESR); no pain + antibodies → silent/postpartum.

BOARD TRAP

Calling a painless antibody-positive thyroiditis 'de Quervain' — de Quervain is painful and antibody-negative.

3. Post-viral trigger

WHAT YOU SEE

Mumps/coxsackie/flu/adenovirus/COVID 'RECENT VIRAL ILLNESS'

WHAT IT ENCODES

De Quervain's classically follows a viral upper respiratory infection by 2–8 weeks — implicated agents include coxsackievirus, mumps, influenza, adenovirus, echovirus, EBV, and (more recently) SARS-CoV-2. There is a strong HLA-B35 association. The history of a preceding viral illness in a patient with a tender goiter is a major diagnostic clue.

4. Very high ESR

WHAT YOU SEE

'ESR ↑↑'

WHAT IT ENCODES

A strikingly elevated ESR (often >50–100 mm/hr), with CRP raised in parallel, is the hallmark lab of subacute thyroiditis and reflects the granulomatous inflammation. A normal ESR makes de Quervain's unlikely. The combination of neck pain + high ESR + low RAIU is the classic triad.

BOARD TRAP

Expecting a normal/mildly raised ESR — a very high ESR is the signature; if ESR is normal reconsider the diagnosis.

5. Low RAIU

WHAT YOU SEE

'RAIU <5% (LOW)'

WHAT IT ENCODES

Radioiodine uptake is LOW (<5%) during the thyrotoxic phase because the disease is DESTRUCTIVE — inflammation releases preformed stored hormone while the damaged gland stops trapping iodine. This is the same low-uptake mechanism in all destructive thyroiditides, and it is the key feature distinguishing thyroiditis from Graves' (where uptake is high).

BOARD TRAP

Expecting high uptake because the patient is thyrotoxic — destructive thyroiditis has LOW uptake; high uptake points to Graves'/toxic nodule instead.

6. Antibodies usually negative

WHAT YOU SEE

'ANTIBODIES usually negative'

WHAT IT ENCODES

In de Quervain's subacute thyroiditis, thyroid autoantibodies (anti-TPO, anti-Tg, TSI) are typically absent or only transiently/low positive — distinguishing it from autoimmune silent/postpartum thyroiditis (TPO+) and from Graves' (TSI+). Thyroglobulin is elevated (released from destroyed follicles) but is not diagnostic.

BOARD TRAP

Expecting anti-TPO to be positive in de Quervain's — antibodies are usually negative; persistent TPO positivity suggests the silent/postpartum (autoimmune) variant.

7. Pain radiates to jaw/ear

WHAT YOU SEE

The seated woman clutching a red, tender neck with a thermometer in her mouth, arrows 'PAIN → JAW/EAR' and 'FEVER' — the exquisitely tender thyroid radiating to jaw/ear with low-grade fever is de Quervain's.

WHAT IT ENCODES

The neck pain of de Quervain's is anterior, exquisitely tender, and radiates to the jaw, ear, or upper chest, often with low-grade fever and malaise. The gland is firm and tender on exam. Pain can migrate from one lobe to the other ('creeping' thyroiditis). It is frequently misattributed to pharyngitis, dental, or ENT pathology.

BOARD TRAP

Misdiagnosing as pharyngitis/otitis/dental pain — the tender thyroid plus elevated ESR redirects to subacute thyroiditis.

8. Granulomas / giant cells

WHAT YOU SEE

'MULTINUCLEATED GIANT CELLS / GRANULOMAS'

WHAT IT ENCODES

Histology of de Quervain's shows granulomatous inflammation with multinucleated giant cells surrounding damaged colloid (the basis of the name 'granulomatous thyroiditis'). Biopsy is rarely needed — diagnosis is clinical (pain + ESR + low RAIU) — but the giant-cell granuloma is a high-yield pathology buzzword.

9. NSAIDs first-line, no ATDs

WHAT YOU SEE

A pharmacy shelf 'NSAID/ASA FIRST · PREDNISONE if severe · β-BLOCKER' over a red banner 'NO ATDs (destruction, not synthesis)' — treat the inflammation, never with thionamides.

WHAT IT ENCODES

Treatment is anti-inflammatory and symptomatic: NSAIDs or high-dose aspirin first-line for pain; a tapering course of prednisone for severe/refractory pain; and a beta-blocker for thyrotoxic symptoms. Crucially NO antithyroid drugs (methimazole/PTU) and NO radioiodine — the gland is not overproducing hormone, it is leaking preformed hormone, so synthesis-blockers and RAI are useless (and RAI can't be taken up by a low-uptake gland).

BOARD TRAP

Giving methimazole/PTU or RAI for thyroiditis — the thyrotoxicosis is from destructive release, not synthesis, so thionamides don't work and RAI uptake is too low.

10. Triphasic course

WHAT YOU SEE

Clock tower 'TRIPHASIC COURSE'

WHAT IT ENCODES

Destructive thyroiditis classically runs a TRIPHASIC course: (1) thyrotoxic phase from preformed-hormone release, (2) transient hypothyroid phase as stores deplete and the gland recovers, then (3) return to euthyroidism. Recognizing the phase explains the labs and prevents over-treatment — the thyrotoxic phase is self-limited and should not be ablated or blocked with thionamides.

11. Phase 1 thyrotoxic

WHAT YOU SEE

'PHASE 1 THYROTOXIC 0–6 wk, release of preformed hormone'

WHAT IT ENCODES

Phase 1 (~0–6 weeks): thyrotoxicosis from leakage of preformed T4/T3 out of inflamed/destroyed follicles. TSH suppressed, free T4/T3 high, RAIU low, Tg high. Treat symptoms with a beta-blocker only — not thionamides. This phase is self-limited as hormone stores run out.

BOARD TRAP

Treating phase 1 with antithyroid drugs — they target synthesis, which is not the problem; only a beta-blocker is needed.

12. Phase 2 hypothyroid

WHAT YOU SEE

'PHASE 2 HYPO 6–12 wk, temp LT4 if prolonged'

WHAT IT ENCODES

Phase 2 (~6–12 weeks): transient hypothyroidism after stored hormone is exhausted and before synthetic recovery — TSH rises, free T4 falls, RAIU may transiently rise during recovery. Most patients are mildly symptomatic and need no treatment; give temporary low-dose levothyroxine only if symptomatic or prolonged, then taper and reassess.

BOARD TRAP

Starting lifelong levothyroxine for the transient hypothyroid phase — most recover; replacement is temporary and should be weaned with TSH monitoring.

13. Postpartum thyroiditis

WHAT YOU SEE

A mother cradling her newborn in the right room, labelled '3–6 MO POSTPARTUM, PAINLESS, RAIU LOW, TPO Ab+' — painless destructive thyroiditis after delivery with positive antibodies.

WHAT IT ENCODES

Postpartum thyroiditis is autoimmune destructive thyroiditis within ~12 months of delivery (or miscarriage), typically thyrotoxic at 1–4 months then hypothyroid at 4–8 months. It is PAINLESS, RAIU is LOW, and anti-TPO is positive (often present in pregnancy). Higher recurrence with future pregnancies and higher risk of eventual permanent hypothyroidism. RAIU and RAI are contraindicated while breastfeeding.

BOARD TRAP

Treating the thyrotoxic phase with antithyroid drugs (or doing a RAI scan in a breastfeeding mother) — it is destructive and self-limited; manage symptoms with a beta-blocker only.

14. ~15% permanent hypo

WHAT YOU SEE

'PHASE 3 RECOVERY; <15% PERMANENT HYPO'

WHAT IT ENCODES

Phase 3 is recovery to euthyroidism in the majority. However a minority — roughly 5–15% of de Quervain's and a higher fraction of postpartum/silent cases — progress to permanent hypothyroidism (higher when TPO-positive), so all patients need follow-up TSH after resolution.

BOARD TRAP

Discharging without follow-up TSH — a meaningful minority become permanently hypothyroid, especially antibody-positive patients.

15. Acute (suppurative) thyroiditis

WHAT YOU SEE

A bottom-corner tag 'ACUTE (BACTERIAL/FUNGAL) — RARE — IV ANTIBIOTICS / DRAIN' — the infected, abscess-forming thyroiditis that needs antibiotics and drainage, not anti-inflammatories.

WHAT IT ENCODES

Acute (suppurative) thyroiditis is a rare bacterial/fungal infection of the gland, usually via a piriform sinus fistula (children) or in the immunocompromised. Presents with severe anterior neck pain, high fever, leukocytosis, and a fluctuant tender mass — but unlike de Quervain's the patient is usually EUTHYROID with NORMAL thyroid function and an abscess on imaging. Treat with IV antibiotics and surgical drainage.

BOARD TRAP

Confusing suppurative thyroiditis with subacute de Quervain's — suppurative is infectious (fever, leukocytosis, euthyroid, abscess → antibiotics/drainage), not a self-limited viral/anti-inflammatory process.